The a_0 Law with Cold Dark Energy Counted Once: the Form the Record Closes On, Its Gates, and What Stays Open

Working note (PAPER72), version 1.2, 2026-11-12 (v1.1: DOI 10.5281/zenodo.24125791). AI-assisted research programme; not peer reviewed; not reproduced by any outside group. Companions: PAPER69 (DOI 10.5281/zenodo.24025170, the closure record), PAPER70 (v2.1, DOI 10.5281/zenodo.24092073, field equations and the record since) and PAPER71 (v1.1, DOI 10.5281/zenodo.24095262, the switch class). $\kappa = \frac{1}{2}$ is fitted. The theory is not closed. Nothing here says the data favour this framework.

Abstract

This note states the form on which the record closes and grades it. It does not complete the theory. The a_0 law alone fails the CMB, the growth of structure, clusters and the Bullet, each for want of the same cold, clustering component of about 5.4 times the baryons. Every way tried of keeping that component out of galaxies has a failing verdict; the last one is recorded here. The form chosen by the owner on 2026-11-09 counts the component once: it is in galaxies too, and around condensed baryons the law describes how it sits. It is real mass, not a second force. In the adopted action this is the static state's own reading: the law's halo is the khronon sector's energy density, carried by the one conserved charge that is dust in the expanding background. We regrade the form with criteria frozen before any number. On groups and clusters it is not falsified, by a gate that can fail from one side only, and the richest clusters sit at its ceiling; the Bullet passes only if the subcluster's stripped gas is counted. On hosted dwarfs it fails: it removes most of the law's shortfall in every class, but two classes stay high and four low rows cannot be reached. On KiDS lensing by galaxy type the test returns no verdict; its readings put early types between the law and the form's truncated halo and cannot place discs; on host-free lenses the form passes a gate that does not separate it from the law. The form passes rotation curves. Its CMB, lensing and growth rows hold when computed, on an amended reading of that lane's checks. With the amplitude of fluctuations pinned by large-angle polarisation, CMB lensing matches the prediction to 0.3σ and the growth rate passes. Raw KiDS weak lensing, on a gravity-only spectrum, prefers an amplitude five of its own widths lower: the absolute fit is excluded at 3σ by the frozen line and reads tension once the reference run's realisation noise is corrected, a check made after grading. This is the tension that any cold component fitted to the CMB carries; the form states neither gas ejection nor neutrino mass, and does not relieve it. With gravity coupled the static state is linearly stable around a point mass; an instability found earlier belonged to a planar background.

Version 1.1 adds the lanes that landed on 2026-11-10 and states the theory in six rules (Section 7). The collapse was computed and breaks down: a clock that starts as dust folds at the centre, and the equations give no answer beyond that moment. The static state is stable around isolated discs away from their centres; a disc in an external field has no verdict, with stability excluded there, and a disc's own centre is open. No dust term tried, keyed on the clock's offset, on its expansion or on its acceleration, joins the falling state to the standing one; the clock's leafwise flux cannot supply the halo; and the action has no cluster state at the form's own dust constant. A survey of media with more structure than one clock finds no class without a failing verdict. On the data side, most of the weak-lensing gap was the survey's superseded release and our own inputs: on KiDS-Legacy, with a neutrino floor, the covariance at the model and the reference run's draw corrected, the form's amplitude sits 1.71/1.88 of the survey's widths high with an acceptable fit, while DES Y3 stays at 2.18/2.23 on a re-scanned fit. These readings rest on post hoc amendments: the lane's amended label is GAP STAYS and its frozen labels are NO VERDICT. With fixed-amplitude reference runs spliced in at low k, an interim reading on one full-size pair puts DES Y3 at 1.85/1.83 and both surveys inside two widths; that lane's frozen label is NO VERDICT, and its amended label, graded after version 1.1 on two full-size pairs, reads AMPLITUDE CONSISTENT, is post hoc and awaits an independent re-grade. The two failing dwarf classes do not survive the record's own systematics budget, and a gas-history reading does not explain them. The form's data rows are computed without the joint between its two states, and several of them fail or have no verdict. The form is not complete.

Version 1.2 adds the lanes that landed after version 1.1 (Section 8). The Lyman-α forest row, computed for the first time at the forest's own redshifts, FAILS on both footings ($\Delta\chi^2$ 9916/9009 against a line of 4); an independent blind build of the same row also FAILS, no a_0 that changes with redshift passes, and one history of the settled radius leaves no window between the forest and galaxy lensing, though a blind re-grade of that lensing gate returns no verdict. No lane finds the joint between the falling and the standing clock, derives κ or the share 5.41, or identifies the medium, and no lane turns a failing gate into a pass; the lanes whose labels read as passes are listed with the words that they await an independent re-grade. We list what is open, the measurement that would decide each item, and our own wrong estimates.

1 Two forms

Cold dark energy is this record's name for a gravitating field component with $w \approx 0$ at the cosmic amount; it is not a particle. Two footings, canonical $a_0 = 9.3603 \times 10^{-11}$ and alt $1.1312 \times 10^{-10}\,\mathrm{m\,s^{-2}}$, are carried throughout and never pooled; pairs of numbers are canonical/alt unless a row says otherwise.

Form A, the law alone, is what PAPER70 states. It passes rotation curves and the Solar System. It fails the CMB (which needs a cold component of 5.41 times the baryons), growth, X-COP by $+0.4675/+0.4345$ dex, the Bullet ($Z = -4.55$, the mass sits on the gas) and Coma's ultra-diffuse galaxies (+1.12 dex with a static external field, +1.41 with a moving host).

Form B, the law with cold dark energy counted once, is the single-counting reading that lanes PD1 and PD2 graded with one fixed share per halo. As graded there it is FAIL/FAIL: it passes its population budget, rotation curves (rms 0.0563/0.0558 against a line of 0.0660) and the carried cosmology, and fails KiDS's pooled outer points (worst -0.635 dex), the groups and the Bullet's amount; X-COP passes only in a reported bracket ($+0.019/+0.010$).

On 2026-11-09 the owner chose B as the form the record closes on, with both forms scored side by side from the record. The choice is of a form, not of a pass: no label was changed by it. It reopens the cold sector retired on 2026-11-01 for this one reading. Section 3 is the regrade that followed.

2 Form B in the adopted action

The action of PAPER70 has certified field equations (lane QX17). A note written after the decision, and checked independently for this paper, reads form B out of them for a static clock.

- **The law's halo is energy density.** The khronon sector's energy density contains $\Pi = D\cdot P$, and the static weak-field 00 equation is Poisson's with source $\rho_b + D\cdot P$. Light and matter both see it.
- **One conserved charge.** The khronon equation is a conserved current whose charge is, to leading order, $D\cdot P + \mu_K^2(Q-1)/K$. With $P = 0$ it is cold dust in the expanding background; around a static source it is the law's halo plus the static dust term $-\mu_K^2\Phi/K$, which is zero at $\mu_K = 0$. Nothing is counted twice.
- **The static state is in force balance.** With $J(g) = (a_0^2/K)[u - \ln(1+u)]$, $u = g/a_0$, the law sector's stress satisfies $\partial_j \Pi_{ij} = (D\cdot P)\, g_i$. In the deep field $J = g^2/2K$, which is the pressure of an isothermal halo with $\sigma^2 = \frac{1}{2}\sqrt{GM_b a_0}$: the relation PD1 had to assume. The identity holds for any $J(g)$ with a curl-free field, so it shows that the action's static state balances itself; it is not evidence for this J.
- **Force balance is not stability; the stability was then computed.** With the metric frozen (the decoupling limit) and in a planar background, lane TPA1 found the action's law-on phase gradient-unstable, with no ghost, wherever the clock's own dust density is below a critical value; the e-folding times on its grid are 5.0 Myr to 0.62 Gyr. The note on form B read the form's static state, which has almost no dust of its own, as lying in that range. Lane SCS1 then coupled gravity around a point mass and labels the state STABLE on both footings: two independent routes find no growing mode at any wavenumber, and disturbances oscillate at 0.08 to 12 times the local free-fall rate. The clock's gradient coefficient is its own charge density, $n_d + D\cdot P$. In TPA1's planar background $D\cdot P = p'$, which is negative; around a point mass $D\cdot P = p' + 2p/r$ is the phantom density, which is positive. SCS1 reproduces TPA1's numbers in the planar background, and the growth returns when that background is put back into the coupled system. TPA1's other findings stand, and the note's sentence has been corrected.
- **What that does not settle.** The result is linear, and for point masses only. For flattened baryons the law's phantom density can be negative in places, and there the same closed form gives growth; SCS1 did not analyse discs or extended spherical distributions. Version 1.1: lane SCS2 analysed discs. Isolated discs (razor-thin exponential and Kuzmin at four masses, a thick disc, a disc with a bulge) are STABLE on both footings: $D\cdot P$ is positive in every cell on both grids and no mode grows on either route. SCS2's frequency known answer fails with its frozen fit and passes under its fourth amendment, which is post hoc and followed a graded run. With a uniform external field along the axis or in the plane the label is NO VERDICT on both footings: both routes find unstable regions, and the magnitudes are not converged or rest on one grid. Lane SCS3 took the external-field case alone and also returns NO VERDICT on both footings; it says that the sign is not in doubt and that STABLE is excluded there. It also says that SCS2's label for isolated discs does not cover the centre, and that whether the static clock is stable at a disc's own centre is open. Lane DCN1 then took the centre. Every reading it froze before its outcome was seen is NO VERDICT, each on a known-answer gate; under the

last of its six post hoc amendments, two of them written after the outcome was seen, it reads ARTEFACT on both footings, with the isolated centre stable. Whatever the reading, every one of its 24 resolved solves has $D \cdot P$ positive in every cell and no growing rate. Extended spherical distributions remain unanalysed. At the dust constant the CMB needs, the static state is Jeans-unstable at the local free-fall rate (e-folding times of 17 Myr to 3.9 Gyr): that is the cosmological setting and not the form's galaxy setting. The lane's three amendments are post hoc and disclosed; one mutation arm does not bite, being a field redefinition, as the lane said before any run.

What the action does not give is stated in Section 5, items 1 to 4.

3 The regrade

Each lane froze its criteria in a commit before any graded number was computed (KO2 ran a model-only design toy before its freeze and says so), and has known answers that must pass and mutation twins that must fail. KO2, RB1, RB2, CO1, CO2, HE1, TPA1 and ONE1 were each re-run by a session other than the one that wrote it, and their labels reproduced; the re-runs are listed in `closure_2026_11_09/RERUNS_2026-11-09.md` and `fable52_push/README.md`. SCS1 (Section 2) landed last: its lead session re-ran both of its routes and recompiled its Lean file (13 theorems); the first route reproduces with only timing lines differing, and the second gives every label identical and every eigenvalue equal to better than one part in a million. The lanes relabel nothing. The lanes added in version 1.1 are in Sections 2, 5 and 7, and Section 7 says which of them were re-run; those added in version 1.2 are in Section 8, which says the same for them.

3.1 Groups, clusters and the Bullet (RB1): NOT FALSIFIED (ONE-SIDED) on both footings

In form B a hot system holds $R = 5.41$ times the baryons that fell into it, and the component stays when gas is expelled or stripped. The retained fraction $f_{\rm ret} = R\,M_{b,\rm obs}/(M_{\rm dyn} - M_{b,\rm obs})$ can therefore not exceed 1 at a system's edge. This gate can fail from above only; the two-sided amount gate stays FAIL as PD1 graded it.

- **Groups.** The 20 hydrostatic groups read $f_{\rm ret} = 0.597$ [0.556, 0.637] at R_{500}, none above 1. Form B reads the missing 40 % as gas outside R_{500}. Nothing on disk measures it.
- **Clusters sit at the ceiling.** X-COP at its last radius reads 1.095 [0.885, 1.259], with 4 of 7 clusters above 1. At R_{500} on the forward-fitted mass the class reads 1.057 [1.046, 1.072], above the ceiling beyond its errors; this row is a reported bracket, and graded there the gate would fail. On the NFW-fit hydrostatic mass the same class reads 0.959. The verdict for the deepest wells turns on about 10 % in the hydrostatic mass.
- **Trend.** $f_{\rm ret}$ rises with depth: slope +0.160 [+0.080, +0.225] in $\log M_{500}$ from groups to X-COP.
- **Radial shape.** One collisionless component with the record's own profile, anchored at each cluster's last radius, gives missing-mass fractions 0.72 / 0.63 / 0.55 / 0.47 / 0.35 per r/R_{500} bin against the measured 0.71 / 0.66 / 0.58 / 0.46 / 0.32 (canonical); bin residuals −0.007 / +0.033 / +0.006 / −0.030 / −0.041 dex against a line of 0.10. This label rests on a post hoc amendment; under the frozen known answer the gate is NO VERDICT.
- **The Bullet.** PD1's 2.6σ deficit in the subcluster is the bookkeeping of the stripped gas. With the gas shared in proportion to the stars the amounts read −0.45 / +0.37 (main / sub) and pass; with nothing reassigned they read +0.49 / −2.58 and fail. No measurement on disk fixes the subcluster's gas before the merger.

3.2 Hosted dwarfs (RB2): B FAILS ON THE HOSTED ROWS on both footings

In form B a dwarf keeps the phantom of its own baryons inside a host, cut only by tides. There is no external field effect.

- **What it removes.** Coma's ultra-diffuse galaxies move from +1.120 to +0.326 dex (alt +0.283) and read B CONSISTENT; the interval contains zero only with the systematics budget, and on statistics alone it is [+0.105, +0.547]. The Milky Way's classical dwarfs read B CONSISTENT as a class (+0.281/+0.241), with Draco, Ursa Minor and Sextans high row by row.
- **Two classes fail high.** The Milky Way's ultra-faints sit at +0.634 [+0.433, +0.865] (alt +0.592) and M31's bright dwarfs at +0.273 [+0.085, +0.473] (alt +0.231). The statistic carries measurement scatter only; a full

systematics budget was not computed, since that would be a change after the freeze.

- **Rows that cut the other way.** NGC 1052-DF2 and DF4 sit 0.74 and 0.94 dex below B and at Newton on their baryons. Tides reach them only if NGC 1052 holds 17 and about 100 times the mass B gives it inside their projected distances. NGC 147 and IC 1613 sit 0.56 and 0.54 dex below B; the law with its external field leaves them as low. Crater II is the one low row tides can reach.

3.3 Lensing by galaxy type (KO2): NO VERDICT for late and early types on both footings

All 181,477 KiDS lenses were fitted with every catalogued neighbour modelled off-centre (888,795 pool objects), in place of a veto. The statistic is the ratio of the amplitude beyond the inner window to the inner amplitude; the law as a galaxy's own field gives 1 and no extended field gives about 0.

- **Why no verdict.** One of 13 known answers failed (a cross-shear pull of 3.11 among 15 amplitudes). Without it the frozen bracket rule gives NOT SEPARATED for both types. One mutation twin did not bite: the ratio measures the shape of the outer profile, not its level.
- **Early types lie between.** They read 0.487 ± 0.071 in the primary design and 0.739 ± 0.115 with lens and neighbour amplitudes untied: below the law's level in every design and above form B's truncated halo (0.038/0.007 in the primary design).
- **Discs cannot be placed.** They read 0.277 ± 0.183 tied and 1.071 ± 0.199 untied. Form B planted with the measured cross-shear as noise reads 1.030 ± 0.282 untied, so the data cannot tell the law from form B for discs.
- **An earlier lead does not generalise.** Lane KO1, and PAPER71 beside its gate, reported that host-free late types show no outer signal (-0.19 ± 0.39 against 1.34 ± 0.18 for early types). Both labelled it a lead and not a result: seen after the freeze and sharing the veto's edge. Plain stacks of all lenses are flat for both types (late outer-to-inner ratio 0.93 ± 0.13, early 1.09 ± 0.08), though they include the neighbours' signal. With neighbours modelled, the deficit appears only inside the host-free subset (-0.03 ± 0.28), which carries the veto's edge.

A look at the host-free lenses, committed before KO2 had a lensing number, compares form B at PD1's share with KO1's shape ratio. It is post hoc and models no neighbours. At the frozen veto radius form B reads 0.510 against a measured 0.607 ± 0.094, with the law as the lens's own field 4.2σ off on the other side. At the widest veto, where the edge effect is weakest, form B reads 0.525 against 0.811 ± 0.134, 2.1σ low, with the law 1.4σ off. The note read from it that one share does not fit late and early types together; ONE1's frozen type-difference line passes at every veto, so the graded statistic does not read that. The same note predicted blind that KO2's late-type ratio would be at most about 0.5; the tied design meets that (0.277 ± 0.183) and the untied one does not (1.071 ± 0.199), so it is not confirmed. Lane ONE1 then graded form B on those lenses with every catalogued neighbour carrying the same share: KiDS host-free PASSES/PASSES by its frozen gate at the widest veto, for PD1's share and for that row only. The shape ratio there is 0.811 against a model of 1.003 (-1.14σ; alt 0.809 against 0.847); late types read 0.084 against 0.792 (-1.46σ) and early types 0.882 against 0.999. At the frozen veto radius, which carries the edge, the model misses (0.607 against 0.924, -2.68σ). The law with no end gives 0.98 at the widest veto, also inside 2σ, so the pass does not separate form B from the law, and the late-type error there (0.42) is too large to test discs. The look's 0.510 to 0.525 was mainly a matter of calibration: with the inner calibration fitted, the model reads 0.86/0.73 at the widest veto without neighbours and 1.00/0.85 with them. Re-run by its lead session: the KiDS script's label and every graded number reproduce, with only timing lines and one status flag differing; its Lean file recompiles (10 theorems) and its twin fails.

3.4 The last way of keeping the component out (HE1): FAILS on both footings

HE1 held the galaxy's share outside a wall as an envelope in pressure balance with the law's field. The envelope exists, is stable and fills 0.2 % of space or less. It fails KiDS's late-type stacks: none of its three stated shares, at any outer radius, passes its first four gates together. Post hoc and with low power, at the widest veto the record's share fits those stacks (p 0.73/0.83), as does the law. Its fifth gate fails as frozen on two clusters near R_{500} only, and the label does not rest on it. Re-run by its lead session: both labels and every number in `he1.json` reproduce; only timing lines differ. Lane TPA1 wrote out and certified the covariant equations of the two-phase action (EQUATIONS COMPLETE; health UNHEALTHY): its switch wall and its dust phase are healthy, and its law-on phase has the instability described in Section 2, which SCS1 finds does not hold around a point mass. With PAPER71's closed switch class, no reading tried that keeps the component out of galaxies passes the record's

gates.

3.5 The cosmology rows, computed (CO1): ROWS HOLD on the amended reading, with an inherited halo list

The decision listed form B's cosmology rows as carried from the cold reference. A further lane computed them. Form B's linear content passes the primary CMB on raw Planck PR4 and ACT DR6 spectra ($-65.21/-65.07$ against a line of 9; cold to baryon 5.410/5.412, fitted). That row replays a fit stored by an earlier lane, and its yardstick is the likelihood package's own test point, not a framework quantity. PR4 lensing reads $0.9617 \pm 0.0231/0.9564 \pm 0.0230$ of the predicted amplitude ($-1.66/-1.89\sigma$ against a 2σ line; shape χ^2 8.56 over 9 bandpowers), and DESI DR1 $f\sigma_8$ passes (largest pull 1.22/1.23). A cold component of the fitted amount gives the cold-component spectra, so these three rows holding is not independent evidence for the framework. For the layout frozen in the criteria (galaxies: the Lagrangian share as the law's phantom; groups and clusters: the share inside R_{200c}, part of it kept in the reference profile) the matter power changes by 0.6/1.0 % at $k = 1$ and 5.1/8.8 % at $k = 3\,h\,\mathrm{Mpc}^{-1}$, and raw KiDS-1000 cannot see it on the 104 points the reference run supports.

- **Two readings of the label.** One mutation arm did not bite as frozen; the label rests on an amendment committed before any graded number. Read as frozen it is NO VERDICT.
- **The halo list is inherited:** a mass function, a bias and a concentration relation calibrated on simulations. The redistribution rows are externally calibrated and say so.
- **The scope decides it.** If a cluster held the whole dust of its turnaround sphere, the power at $k = 1$ would be 2.9 times higher and raw KiDS would reject it. The form's cosmology needs the cluster share tied to the baryons present (5.41 times them) and counted inside R_{200c}; RB1 grades the same bookkeeping at R_{500} or the last measured radius.
- **CO1 left the amplitude open.** The fit's σ_8 is 0.8405/0.8429, with no low-multipole data to hold it. Raw KiDS fits the record's reference run at 0.811 with $p = 0.014$ and form B at its own amplitude with $p = 0.0015/0.0009$ (a bracket with constant-factor scaling, which understates it). The forest row stays carried. Version 1.2: the forest row has since been computed at the forest's own redshifts and FAILS on both footings, and on the record's own halo list the redistribution rows are NO VERDICT as frozen (Section 8.1). CO2, below, pins the amplitude and finds that these absolute p-values carry the reference run's realisation noise.

3.6 The amplitude, pinned (CO2): AMPLITUDE EXCLUDED BY RAW KiDS AT 3 SIGMA as frozen, on both footings; not robust

Lane CO2 refitted the same spectra with the Planck 2018 low-multipole polarisation table added. The optical depth moves from 0.0697/0.0729 to 0.0532 ± 0.0066 on both footings (a width along the degeneracy only, which understates the profile width) and σ_8 from 0.8405/0.8429 to 0.8271; the cold to baryon ratio stays at 5.411/5.412.

- **The CMB side closes.** The pinned fit is adequate ($-63.79/-63.75$ against a line of 9). PR4 lensing reads $0.9928 \pm 0.0239/0.9929 \pm 0.0239$ of the prediction (-0.30σ on both footings, from $-1.66/-1.89$ unpinned), and DESI DR1 $f\sigma_8$ passes (χ^2 1.95 over four bins).
- **Weak lensing is where it fails.** For form B's spectrum raw KiDS-1000 is best at $\sigma_8 = 0.7358 \pm 0.0162/0.7341 \pm 0.0161$, and the pinned value sits 5.62/5.78 widths above it; with one 0.06 eV neutrino (a bracket at the pinned parameters, not a refit) σ_8 is 0.8152 and the gap is 4.89/5.04 widths. The widths are KiDS's alone. Across every variant below the gap is 5.1 to 5.8 widths, and the lane calls it its robust statement.
- **The label is at its line.** The graded row is an absolute fit. Raw KiDS at the pinned amplitude, on the record's gravity-only run moved by the run's own response, reads χ^2 157.89/159.35 on 104 points, $p = 0.0026/0.0021$, just below the frozen 3σ line of 0.0027. With a constant-factor move (a bracket) the same row reads $p = 0.0044/0.0037$, tension and not exclusion; the reference run alone at the same amplitude reads χ^2 156.88/156.85; with the neutrino bracket $p = 0.0090/0.0077$. Unpinned, the same row reads $p = 0.0004/0.0002$, so the pin moves the form towards the lensing data and not far enough. DES Y3 through the same path reads $p = 0.050/0.051$ (reported, not graded).
- **The exclusion does not survive a check made after grading.** The reference run is a single box of $100\,\mathrm{Mpc}/h$ whose large-scale power is a high draw: its starting spectrum is 1.48 times its own linear spectrum in its third bin. A difference of two models on the same run cancels this; an absolute fit does not. With the realisation taken out in four ways (the run's own starting ratio with two tapers, and two $200\,\mathrm{Mpc}/h$ runs below

$k = 0.5\,h\,\mathrm{Mpc}^{-1}$), p is 0.003 to 0.013: tension each time. The check is post hoc, is reported and changes no label; the lane says it would not quote the exclusion without it. CO1's absolute KiDS p-values carry the same noise (its reference row, $p = 0.014$, reads 0.021 to 0.049 corrected); CO1's label rests on a difference and is not affected.

- **What the label covers.** Form B as stated, a gravity-only run with nothing ejecting gas beyond R_{200c}, and the inherited halo list; the fit keeps massless neutrinos and the shear row uses 104 of the 225 published KiDS points. The lane's criteria say, before any number, that a fail here is the weak-lensing amplitude tension that any cold-component fit to the CMB carries: it is not peculiar to form B, and form B does not relieve it. The two lines (0.05 and 3σ) are conventional but were chosen with CO1's numbers in view, which the lane states as a limit. Its one amendment, a known-answer tolerance, is post hoc and was committed before any graded number; the known answer passes the frozen tolerance too.

4 Form B after the regrade

Gate	Reading for form B	Lane
Rotation curves	pass, rms 0.0563/0.0558 (line 0.0660)	PD1
Solar System	pass, carried	PD1
CMB peaks, CMB lensing, growth rate	hold when computed; NO VERDICT with one mutation arm as frozen; redistribution rows on an inherited halo list; on the record's own halo list NO VERDICT as frozen, rows hold under a post hoc amendment that awaits an independent re-grade (version 1.2)	F1, CO1, HLF1
Forest (version 1.2)	FAILS at the forest's own redshifts, $\Delta\chi^2$ 9916/9009 against a line of 4; an independent blind build FAILS (8142/7686); no $a_0(z)$ passes; with the law's layout only inside a settled radius FAILS, FRAGILE; one settled-radius history WINDOW EMPTY; flux level [PENDING: FRZ3, the forest row at flux level, is running]	FRZ1, FRZ1R, FRZ2, FZS1, SRH1
Amplitude of fluctuations, pinned	CMB lensing (-0.30σ) and growth pass; raw KiDS prefers σ_8 0.736 against 0.827, 5.1 to 5.8 of its widths; absolute fit excluded at 3σ as frozen ($p = 0.0026/0.0021$), tension once the reference run's realisation is corrected	CO2
Weak-lensing amplitude on current releases (versions 1.1, 1.2)	KiDS-Legacy and DES Y3: NO VERDICT as frozen, GAP STAYS post hoc (KL1); with fixed-amplitude reference runs NO VERDICT as frozen, AMPLITUDE CONSISTENT under a post hoc amendment that awaits an independent re-grade (BX1); on DESI DR1 lenses NO VERDICT as frozen, AMPLITUDE CONSISTENT, FRAGILE post hoc, and the same word, FRAGILE, from a blind re-grade that says it is not fully blind, with DES Y3 and KiDS-1000 plus DES Y3 in TENSION (LWB1, LWB1R); a consistency check, not a pass	KL1, BX1, LWB1, LWB1R
Clusters, amount	not falsified (one-sided); at the ceiling; two-sided gate FAIL; what the non-baryonic mass is: NO CANDIDATE (version 1.2)	RB1, PD1, CLS2
Clusters, radial shape	consistent under a post hoc amendment; NO VERDICT as frozen	RB1
Groups	not falsified (one-sided), f_{ret} 0.597; two-sided gate FAIL; the census to turnaround NOT EVALUABLE on the lensing on disk and on DESI groups	RB1, PD1, GO1, GO2
Bullet	consistent only if the subcluster's stripped gas is counted	RB1
Coma UDGs	consistent on the systematics budget	RB2
Milky Way classical dwarfs	consistent as a class	RB2
Ultra-faints, M31 bright dwarfs	FAIL high	RB2
DF2, DF4, NGC 147, IC 1613	against the form; tides cannot reach them	RB2
KiDS lensing by type	NO VERDICT; early types between the law and the form	KO2
KiDS host-free lenses	passes at the widest veto with neighbours carrying the share; does not separate the form from the law	ONE1
Outer cluster shear	FAIL, about 3σ with systematics	ZN4, FR1

Gate	Reading for form B	Lane
Static state reached by collapse	BREAKS DOWN (version 1.1): the falling clock folds at the centre and the equations give no answer beyond	COL1
Stability of the static state	linearly stable around a point mass and, away from the centre, around isolated discs with gravity coupled; a disc in an external field NO VERDICT with STABLE excluded; the earlier instability belonged to a planar background; its nonlinear outcome NO VERDICT (version 1.2)	SCS1, SCS2, SCS3, TPA1, EFN1
One dust term for the CMB, clusters and galaxies (version 1.1)	NO VALUE for a quadratic and for an exponential term; keyed on the expansion BREAKS AT TURNAROUND; keyed on the acceleration BREAKS AT A1; gated on the halo density NO VALUE (version 1.2)	MK1, MK2, TK1, AKD1, PIK1
The joint (version 1.2)	not found: every lane that grades a carrier or a settling route fails, has no verdict, or reads otherwise only under a post hoc amendment that awaits an independent re-grade (Section 8.2)	EDG1, SET1, VSH1, VSJ1, AKS1, CLOSE2, ...
Milky Way, DESI DR2 and the literature (version 1.2)	the enclosed-mass profile passes and the extrapolated M_{200c} fails (DW1); the line passes the shape-measuring curves (NMB1, FRAGILE in one case) and the outer masses (PLT1, FRAGILE; satellites beyond 150 kpc FAIL LOW, post hoc); hot gas does not supply the rotation curve (MWG1); each pass awaits an independent re-grade	DW1, NMB1, PLT1, MWG1, MWJ1
Euclid strong lenses (version 1.2)	the law on the stars alone: NO VERDICT on the full sample, at its ceiling; WIDE LENSES BREAK THE CEILING, confirmed by a blind re-grade; source redshifts and dispersions NOT EVALUABLE	EUC1, EUC4, EUC4R, EUC2, EUC5
Flat a_0 at $z = 2.33$ (version 1.2)	NO VERDICT as frozen; FLAT HOLDS / EXPANSION-MODEL DEPENDENT under a post hoc amendment; does not separate the framework from ΛCDM; a blind re-grade reads the same (FLAT FAILS at Ω_m 0.25 and 0.35)	AZ2, AZ2R

5 What stays open

1. **The collapse.** For a static clock the action's equations give the law's shape and hold it. No solution takes a free-falling clock, dust falling in with the baryons, to that state. This is the spiral wall of the 2026-11-01 ledger. Version 1.1: lane COL1 computed it and labels it BREAKS DOWN on both footings. A clock that starts as cold dust in the Hubble flow at $z = 10$ around a baryon point mass folds at the centre within 0.063 to 0.154 Gyr, at the three dust constants its brief asked for and at all four masses, and the equations give no answer beyond that moment; one of its known answers fails, so its law-off rows and one bracket are NO VERDICT. Lane FLD1 (DOES NOT on both footings) finds that the folded region today holds every SPARC galaxy's last measured radius (175 of 175) and every galaxy of the high-redshift sample on disk at its stated extent (100 of 100), and stops 1.3 to 5.4 times short of the KiDS windows and 1.1 to 4.2 times short of the form's own cap (canonical). Lane REL1 tried the law on the clock's acceleration relative to the baryons: NOT WELL-DEFINED, since matter's equation of motion becomes fourth order. Lane LFS1 asked whether the clock's leafwise flux can supply the halo: ILL-POSED, and the answer is no; the flux adds no charge. Version 1.2: no route tried reaches the static state. Settling by relaxation towards the law's pressure STRUCTURE FAILS AT A1 (SET1); no local dissipation of the clock closes the joint (AKS1, a post hoc part); a viscous shock settles the clock only under a post hoc amendment (VSH1, which awaits an independent re-grade), and that settled state is not the joint (VSJ1, also post hoc); an engineered kinetic function FOLDS (CLOSE-1); and an edge layer cannot hold the charge (EDG1, FAILS AT E3).
2. **Stability beyond a point mass.** With gravity coupled the static state is linearly stable around a point mass. Version 1.1: isolated discs are stable too away from their centres (SCS2). A disc in a uniform external field, where the law's phantom density is negative in places, has no verdict (SCS2, SCS3), and SCS3 says that STABLE is excluded. A disc's own centre is open, and the nonlinear outcome has not been computed. Version 1.2: lane EFN1 took the nonlinear outcome in an external field and reads NO VERDICT on both footings in every reading: its direct route stops at an artefact of its own boundary, and its weakly nonlinear route is not converged.

3. **One dust term.** With the quadratic $K(Q)$ the background has $w = w_0/(w_0 + a^3)$ with $w_0 = I_0/4\mu_K^2$. Our arithmetic, not a lane, with H_0 70 and Ω_c 0.26: $w_0 = 1.063 \times 10^{-8}/\mu_K^2$ (μ_K per Mpc), so at $\mu_K = 3.71$ per Mpc the component still has $w = \frac{1}{2}$ at $z = 1089$. Dust by recombination needs μ_K well above that; the galaxy-lensing bound is 0.1. Version 1.1: lane MK1 labels the question NO VALUE on both footings. The CMB needs μ_K of at least 198 per Mpc for the component to be cold by $z = 3400$ and 2.51×10^4 for its lensing shape, far above anything the galaxy rows allow [PENDING: an uncommitted correction notice in fable52_push/MK1_one_dust_constant/RESULT.md bears on the 2.51×10^4 threshold; the committed wording is kept until it is committed or withdrawn]. MK1 also corrects the bound quoted above: 0.1 is the first failing grid value, and the bound lies between 0.03 and 0.1. Lane MK2 returns NO VALUE for an exponential dust term (as frozen NO VALUE/NO VERDICT). A stiffness keyed on the clock's expansion BREAKS AT TURNAROUND (TK1), and one that falls with the clock's acceleration BREAKS AT A1 (AKD1); Section 7 gives both. Version 1.2: a dust term gated on the law's own halo density (PIK1) reads NO VALUE on both footings; its file says the verdict is carried by an analytic derivation evaluated on the record's published numbers, not by a fresh numeric pipeline, and that a re-run is owed [PENDING: an uncommitted correction notice in fable52_push/PIK1_pi_gated_dust/RESULT.md bears on this lane's verdict; the committed wording is kept until it is committed or withdrawn].
4. **Clusters are not a static-clock state.** Closing X-COP with the static dust term needs μ_K of 1.78/1.74 per Mpc, 17 to 18 times the galaxy bound. No other state has been solved. Version 1.1: MK1 corrects the sentence before: 1.78/1.74 is only where the mean residual is zero, the shape passes at no μ_K, so the dust does not close X-COP at any value, and '17 to 18 times' should read '18 to 60 times'; done as the action's own static equation the mean-zero value is 1.53/1.46. Lane CLS1 searched the action for a cluster state. As frozen its label is STATE EXISTS-PARTIAL, with 0 of 7 clusters holding a state; at the form's own dust constant (198.25 per Mpc and above) it is NO STATE on both footings, proven on all seven. Version 1.2: lane CLS2 asked what the clusters' non-baryonic mass is under form B and reads NO CANDIDATE on both footings: three candidates are killed by proof and the fourth, gas that is under-counted, fails on a budget.
5. **Groups** are untested from below: form B needs 40 % of their baryons to lie outside R_{500}. Version 1.1: lane GO1 tried the census in group outskirts and is NOT EVALUABLE: the lensing on disk does not reach the turnaround radius. Version 1.2: lane GO2 repeated it on isolated DESI groups and is again NOT EVALUABLE on both footings; where the turnaround radius can be located at all, the error on $f_{\rm ret}$ is 0.9 to 1.8 against a gate of 0.20.
6. **The richest clusters** sit at or above the form's ceiling, depending on the hydrostatic mass used. Version 1.1: lane XCV1 regraded X-COP with a mass that does not assume hydrostatic equilibrium: NO VERDICT as frozen, since a known answer fails; under its post hoc amendment the class is AT THE CEILING with the central value above it.
7. **The Bullet** passes only with gas credited to the subcluster that no measurement fixes.
8. **Ultra-faints and M31's bright dwarfs** sit above the form. Version 1.1: with the record's systematics budget and the dispersion upper limits counted, neither class's interval excludes zero (DG1); the medians stay high. A blind regrade with a full budget (DSB1) reads NO POWER for every class under its amended reading. Version 1.2: three readings that would put the ultra-faints low (stripping, an external field registered at formation, falling in at the host's field) each FAIL THE PAIR with X-COP on both footings (CRACK-1): each predicts them low, and they sit at +0.592/+0.496.
9. **DF2 and DF4** sit below it and need a host 17 to 100 times heavier than the form gives. Version 1.1: lane NH2 set NGC 1052's measured mass against that need: NO VERDICT as frozen, since two known answers fail; under its post hoc reading tides cannot reach DF4 and DF2 is not decided.
10. **KiDS.** On all lenses, early types need more beyond the inner window than PD1's truncated halo and less than the law. On host-free lenses the form passes a gate that the law passes too. Discs are not tested by either. Version 1.2: a blind re-grade of SRH1's lensing gate (SRH1R) finds that the gate passes equally at a settled radius 0.05 of the fold and with the cold reference everywhere, so this lensing test cannot tell one settled radius from another.
11. **The outer cluster shear** stands near 3σ.
12. **Cosmology.** The weak-lensing amplitude is not solved: the CMB pins σ_8 at 0.8271 and raw KiDS prefers 0.736 for the form's spectrum, 5.1 to 5.8 of its widths lower. Form B states neither gas expelled from haloes nor neutrino mass. The forest row is carried, and the redistribution rows rest on an inherited halo list and on the cluster share being counted inside R_{200c}. Version 1.1: lane KL1 took the gap apart (Section 7). On the surveys' current data the gap is 1.71/1.88 widths on KiDS-Legacy, with an acceptable fit, and 2.18/2.23 on DES Y3 (2.89/2.82 on the frozen scan). These are post hoc readings: the amended label is GAP STAYS

and the frozen labels are NO VERDICT. Lane BX1's interim reading with fixed-amplitude reference runs puts DES Y3 at 1.85/1.83; its frozen label is NO VERDICT. Version 1.2: the forest row was computed and FAILS (Section 8.1). BX1's amended label, graded on two full-size pairs, is AMPLITUDE CONSISTENT on both footings, with DES Y3 at +1.78/+1.76 widths; it is post hoc and awaits an independent re-grade. On DESI DR1 lenses LWB1 and its blind re-grade read AMPLITUDE CONSISTENT, FRAGILE (Section 8.4). These are consistency readings and close nothing; the item stays open.

6 What would decide it

- The spherical collapse of the clock around a baryon mass with the record's own equations was computed in version 1.1 and breaks down (COL1). What would decide the joint now is an equation for what the clock is behind a fold; Section 7 lists what the landed verdicts force on it.
- The stability of the static state for a disc in an external field with converged magnitudes and for an extended baryon distribution, and its nonlinear outcome. Point masses and isolated discs away from their centres are computed and stable.
- The baryons between R_{500} and the turnaround radius around the same groups (thermal SZ, X-ray, stars) with a lensing mass to that radius. Form B needs $f_{\rm ret} = 1$ at turnaround; a census that falls short falsifies it.
- A cluster mass beyond R_{500} that does not assume hydrostatic equilibrium.
- A measured mass for NGC 1052 inside 79 and 165 kpc.
- The dwarf classes regraded with a full systematics budget, frozen before the numbers. Version 1.1: lane DSB1 did this blind. Its frozen reading STOPPED at a known answer and has no label. Its amended reading, with that one known answer replaced after the freeze and before any graded number, is NO POWER for every graded class on both footings: no class fails, every median stays above zero, and the smallest offset each class could detect (0.59 to 1.23 dex) is above the frozen 0.30 dex line, so the classes cannot decide the form either way. The Milky Way's ultra-faints read +0.572 [−0.043, +1.173] (alt +0.542 [−0.087, +1.127]); reported and not graded, they read as failing high if their nine upper limits are fixed at the limits or dropped. What would decide the dwarf rows now is data with more power, not another regrade.
- Lensing by type with more area. A spectroscopic veto with the redshifts on disk has no power: the late-type error at the widest veto is 0.68/0.62 of the law in KO1 and is forecast at 0.57/0.52 with such a veto, against about 0.47 needed.
- For the weak-lensing amplitude, now that the CMB side is pinned: a refit that carries neutrino mass, in place of the bracket at fixed parameters; the shear row on more than one reference run, and on all of the published points; and a statement by the form of whether gas leaves haloes. Version 1.1: KL1 made the refit and put the measured gas in; it finds the single reference run to be the limiting input, so the deciding step is the shear row on reference runs without realisation noise. Lane BX1 has done this for one full-size pair (Section 7); its second pair was still running when version 1.1 was written and has since finished (Section 8.4). Version 1.2: for the forest row, lane FRZ3 is repeating the test at flux level and had not landed (see the gate table); for the strong lenses, the source redshifts of Euclid's lenses, for which EUC1 froze a gate and which EUC2 could not find in DESI's fibres; for the groups, a stack that reaches the turnaround radius, which GO2 did not.
- Gaia DR4 (2026-12-02), scored as pre-registered.

7 Version 1.1: the theory in six rules, the rows since, and what a completion must satisfy

The lanes in this section landed on 2026-11-10, after version 1.0, except the earlier lanes named in the survey of media. Each lane froze its criteria first; where a frozen reading and a post hoc reading differ, both are given, the frozen one first. The closure session re-ran DG1's main run (not its mutation twin), all three parts of KL1 and AKD1 from copies, and every label and stored number reproduced apart from run times and memory (`closure_2026_11_10/RERUNS_2026-11-10.md`); a re-run reproduces a lane's arithmetic and is not an independent derivation. KL1's second part was also regraded by lane KL1R, with the same labels. The other lanes named here are quoted from their result files and were not re-run for this note; for the lanes under `fable52_push/`, what their lead session checked is recorded in that folder's `README.md`.

7.1 The theory in six rules

1. **Fields.** The metric, one clock field that picks a slicing, and a polarization on its slices. Matter couples to the metric only. The action is lane QX17's action (A), which is PAPER70's.
2. **One conserved charge.** The clock equation is a conserved current. The universe carries 5.41 units of its charge per unit of baryons, a fitted number. It is not derived: in the action it is an integration constant (note R541, which is not a graded lane), and four attempts to derive it are EXCLUDED on both footings: a clock-matter coupling (R541A), a Gauss-Bonnet source (R541C), a baryon-current coupling (R541D) and a coupling to $B - L$ (R541E). A fifth, an early potential on the clock (R541B), reads TRADES: it moves the number into the potential's height and does not fix it.
3. **Falling clock.** Where the clock free-falls with the expansion the charge is cold dust. This is the cold component of cosmology.
4. **Standing clock.** Where the clock stands still around baryons its own equations give the law, with $a_0 = \frac{1}{2}c\sqrt{G\rho_{\rm DE}}$ and $\kappa = \frac{1}{2}$ fitted. The halo is the charge itself and gravitates and lenses as mass.
5. **Counted once.** A halo is made of the charge its region brought. A hot system holds 5.41 times the baryons that fell into it. A body that brought no charge has no halo.
6. **The joint.** How a falling clock becomes a standing one is not derived.

The data rows of this note are computed from rules 3 to 5, and several of them fail or have no verdict. Rule 6 is the missing equation, and the hot branch of rule 5 is a rule and not a solution of the action.

7.2 What the landed verdicts force on the joint

Any completion has to meet all of these. Each line rests on a lane's verdict on both footings; where a sentence is a reported reading and not a label, it says so.

1. **Not a wall at a fixed place.** HE1 FAILS, and PAPER71's switch class is closed.
2. **Not a dust term that depends on the clock's offset alone.** MK1 and MK2 return NO VALUE (Section 5, item 3). The offset is about 10^{-6} both in the background at $z = 3400$ and in a galaxy's well today.
3. **Not the expansion alone.** Lane TK1 keyed the dust stiffness on the clock's expansion: BREAKS AT TURNAROUND. The key keeps the expanding background well posed, with a margin of 1.19 on the Jeans line. A collapsing region passes through small expansion while its clock is still falling; there short waves grow without bound, 2 to 140 Myr before the turnaround, and the region never turns around.
4. **Not a stiffness that falls with the clock's acceleration.** Lane AKD1: BREAKS AT A1. The key is a negative copy of the law's own term. Its weight against the law is $3.9 \times 10^{14}/4.2 \times 10^{14}$ at $z = 3400$ and $4.5 \times 10^{11}/4.9 \times 10^{11}$ at $z = 1100$, so the clock's inertia is negative on every sub-horizon scale there, until $z = 11.6/11.4$. A shape that is flat at zero acceleration has the same sign problem in the outskirts of static wells; that is a reported class bound and not a label.
5. **Not the law on the clock's acceleration relative to the baryons.** REL1: NOT WELL-DEFINED.
6. **It has to survive the fold.** COL1: BREAKS DOWN. Nothing in the action removes the infall energy.
7. **It has to reach further than today's fold.** FLD1: DOES NOT.
8. **It cannot draw the halo from the clock's leafwise flux, and it needs a cluster state the action does not have.** LFS1: ILL-POSED. CLS1: NO STATE at the form's own dust constant.

So no dust term tried that is built from the clock's own first-order kinematics, its offset, its expansion or its acceleration, joins the two states. Of functions of the offset, only a quadratic and an exponential were graded; MK1's statement about any function of the offset alone is a requirement and not a graded result. One kinetic function with a narrow interval of changed compressibility is named in the two assembly notes: its linear and static checks pass there, and whether a forming halo can reach its static state is not computed. It is not a result.

7.3 Media with more structure than one clock

A design note, `closure_2026_11_10/MEDIUM_DESIGN_2026-11-10.md`, which is not a lane, sets the classes of medium the record has tried against four properties the data ask for: conserved and cold; passing through itself; hot and at full share in clusters; locked to the baryons in galaxies. Each class has a failing verdict in its own lane: any local, momentum-conserving, normally self-gravitating medium whose equilibrium is the law (M5: NO-GO, a theorem); two phases (PH1: NO-GO); a superfluid with phonons (QX28: FAILS at G3, G4, G5, G6 and G8); a self-interacting isothermal inner halo (IJ1: FAILS); a light fermion beside the law (HF1: fails its first gate at

5 of 6 masses with 1 undetermined; the lane says this kills the family on its inputs and is not a no-go theorem) and a warm component beside it (WM1: FAIL AT SEEDING, a gate added after its first result, which failed on groups); the tethered medium (TM1: it does not resolve law against clumping, and its scored rows fail or are conditional); and nonlocal, history and global actions (FA3: no candidate survives). Route E, in which cold dark energy clumps everywhere, gravity is Newtonian and the law is emergent, had its cold sector retired by the owner on 2026-11-01.

One class had no lane: a medium in thermal contact with the baryons where their acceleration exceeds a_0. The note works it out as our own arithmetic, not a graded result. The contact needs an opacity $\pi G/a_0$ of $22.4/18.5\,\mathrm{cm^2\,g^{-1}}$. With it the medium's mean free path against baryons at $z = 1100$ is 25.8/31.1 pc and its drag at a relative speed of $30\,\mathrm{km\,s^{-1}}$ is 0.73/0.61 of the expansion rate, so it does not stay decoupled from the plasma; and a cluster's gas is 0.14 to 0.68 optical depths (canonical), so it would be dragged with the Bullet's gas.

Three lanes then took the question as an inverse problem. Lane PAR1 asked which parent action could carry the cold dark energy at all: NO VERDICT on both footings, because two branches stayed open with no member of them written and graded there. Its statement is that no parent action built from one single-valued carrier exists, a theorem that rests on a post hoc amendment, so a working parent would have to carry the cold dark energy on a multi-valued field. It also reproduces the three numbers of the arithmetic above independently. The two open branches were then graded. Lane WV1, wave dust with a split charge: one variant reads STRUCTURE FAILS as frozen; the other is NO VERDICT as frozen and STRUCTURE FAILS under two post hoc amendments. Some of its runs were paused when version 1.1 was written; they finished on 2026-11-11 and add a failing part, a resolution failure at the larger mass, and under the first amendment alone that variant's reading moved from NO VERDICT to STRUCTURE FAILS, which the lane says rests on a resolution failure, not on physics (version 1.2). Lane SMF1, the simplest smooth-frame parent: STRUCTURE FAILS on both footings at its first part. In any region bounded where the frame free-falls, the frame's polarization charge is zero at every time, so the exchange cannot fill the law's halo in net. A post hoc amendment adds that neither failing gate is specific to waves, so a collisionless carrier in the same structure does not escape either one.

7.4 The data rows since version 1.0

- **Weak-lensing amplitude (KL1).** CO2's gap of 5.62/5.78 KiDS-1000 widths came apart into pieces. The largest is the survey's own release: for the form's spectrum KiDS-Legacy prefers $\sigma_8 = 0.7767/0.7748$ where KiDS-1000 preferred 0.7358/0.7341. The largest on our side is the massless neutrinos: a refit of the CMB fit with one 0.06 eV neutrino gives $\sigma_8 = 0.8121/0.8120$. The covariance evaluated at the model and the gas RB1 measured outside R_{500} are smaller pieces. As frozen, the lane's first two parts read NO VERDICT, because a mutation arm cannot bite on DES Y3. Under a post hoc amendment, KiDS-1000 with DES Y3 reads TENSION (KiDS-1000 gap 3.95/4.06 widths) and KiDS-Legacy with DES Y3 reads AMPLITUDE CONSISTENT WITH RAW LENSING ($p = 0.111/0.101$ for Legacy), with Legacy still 2.13/2.30 of its widths below the form, which the lane's frozen list counts against the form. Lane KL1R, a regrade of that part by another session, returns the same two readings; it says its blindness was not clean, since KL1's headline numbers were visible before it began, and that it is not an independent check of the shared machinery.
- **The second pass.** With the reference run's known draw corrected and the cold component responding to the moved gas, the lane reads NO VERDICT as frozen (a mutation arm does not bite, and on alt a known answer fails on DES Y3) and, under three post hoc amendments, GAP STAYS, driven by DES Y3: KiDS-Legacy 1.71/1.88 widths and consistent (2.06/2.23 with the full Gaussian likelihood), DES Y3 2.18/2.23 on the polished scan (2.89/2.82 on the frozen scan), KiDS-1000 3.72/3.85. Held to Legacy's own alignment model under its own prior, Legacy reads 1.57/1.83 and stays consistent. Gas taken from a stellar-mass-selected stack is a bracket and barely moves anything. The lane concludes that the remaining gap is the size of the reference run's realisation noise: two independent $200\,\mathrm{Mpc}/h$ runs spliced in at low k put DES Y3 at 1.31/1.27 against 3.49/3.43 widths. Lane BX1 then replaced those single runs below $k = 0.5\,h\,\mathrm{Mpc^{-1}}$ with a fixed-amplitude run and its inverted-phase twin in a $200\,\mathrm{Mpc}/h$ box. As frozen its label is NO VERDICT on both footings: one known answer fails and one mutation arm does not bite. Under its post hoc amendment the graded label is the mean of two full-size pairs and was still pending when version 1.1 was written, because the second pair had not finished; it landed afterwards and is given in Section 8.4. Its interim reading on the one full-size pair is AMPLITUDE CONSISTENT on both footings: KiDS-Legacy +1.51/+1.72 widths (p 0.26/0.25) and DES Y3 +1.85/+1.83 on the polished scan (p 0.30/0.31), with KiDS-1000 at +3.59/+3.72. The lane puts a single pair's residual noise at about 0.1 width on Legacy and about 0.2 width on DES Y3, so the DES reading is

inside that noise of the two-width line; with the full Gaussian likelihood DES Y3 is above 2 on both footings (+2.27/+2.18). The closure session re-ran the lane's known answers, splice, re-grade and grading from the stored spectra, and both readings and every gate number reproduce; the box simulations themselves were not re-run.

- **Hosted dwarfs (DG1): HISTORY DOES NOT CLOSE THEM on both footings.** RB2's two failing classes rested on measurement scatter alone. With the record's joint systematics budget re-derived for the form and the dispersion upper limits counted, both intervals contain zero on both footings: the ultra-faints read +0.550 [−0.090, +1.175] (alt +0.509 [−0.110, +1.141]) and M31's bright dwarfs +0.299 [−0.181, +0.774] (alt +0.258 [−0.237, +0.747]). The medians stay high; this is the standing RB2 gave Coma's ultra-diffuse galaxies and not a pass. A gas-history reading, in which a dwarf keeps the halo of the gas it lost, brings the class medians to −0.045, +0.037 and −0.010 dex and fails the per-galaxy gates: inside the ultra-faints the rank correlation of the residual with the predicted history is +0.06, and a two-number stellar-mass power law fits better (rms 0.306 against 0.343 dex). The lane is post hoc to RB2's class numbers and says so.
- **The cold component's shape (SHP1).** Its label is THE LAW'S SHAPE ON ROTATION CURVES, LENSING ABOVE; NOT ONE SHAPE (R3b) on both footings. On rotation curves (2700 points in 149 galaxies) everything that is not baryons follows the law's shape in every bin at the standard mass-to-light ratios, the worst bin median within 0.057 dex; at disc ratios of 0.3 or 0.8 that pass fails. KiDS isolated lenses have the same shape at $1.40 \pm 0.03/1.28 \pm 0.03$ times the law with the published stars only, and sit below it with the published hot gas counted as baryons. A separate halo of PD1's 59.5 M_b on top of the law is excluded on every reading.

8 Version 1.2: the verdicts since version 1.1

The lanes in this section landed after version 1.1 was drafted on 2026-11-10; SET1, NBC1 and the five lanes of the TOE battery landed that day and were not in it. Each froze its criteria first unless its row says otherwise. Frozen readings come first and post hoc readings are labelled. A lane whose label reads as a pass, a derivation or a rescue awaits an independent re-grade unless a blind re-grade is named; none turns a gate from fail to pass or closes an open item. A re-run is named only where the lane's file records it, and a re-run reproduces arithmetic; it is not an independent re-grade. An independent audit of 2026-11-12 graded many of these lanes (`reviews/hy4_codex_audit_2026_11_12/`, files A1 to A4: SOUND, SOUND WITH LIMITS, OVERSTATED or WRONG); its grade stands beside the lane's own label. Its plain answers: κ is derived nowhere, and neither is 5.41; the joint is not found and the medium is not identified; no lane in its data slice rescues a failing row. Where another session has left an uncommitted correction notice in a lane's file, this note keeps the committed wording and marks the sentence as pending.

8.1 The forest

Version 1.1 carried the forest row from the cold reference, and CO1's redistribution rows used an inherited halo list at $z = 0$. Lane HLF1 rebuilt them on the record's own halo list: NO VERDICT on both footings. Its Part A is NO VERDICT as frozen, since a mutation arm did not bite, and reads ROWS HOLD WITH THE OWN LIST under a post hoc amendment, which awaits an independent re-grade. Its Part B found that the forest gate fails when CO1's $z = 0$ redistribution is held to the forest's redshifts (160.7/1025.6) and passes trivially under the record's fade, so the row turned on how the redistribution evolves to $z \approx 3$.

Lane FRZ1 computed that evolution: FOREST ROW FAILS on both footings, $\Delta\chi^2$ 9916/9009 against the record's gate of 4. Because the law's physical scale is fixed, galaxies' cold dark energy is laid out flatter at the forest's redshifts than in the cold reference, and the power falls: at $z = 3$ the ratio to the cold reference at k 3/5 $h\,\mathrm{Mpc}^{-1}$ is 0.845/0.665 (canonical) and 0.831/0.669 (alt). Every frozen bracket fails too, so the label carries no FRAGILE. Its lead re-ran it from a copy and the outputs reproduce; the file says this is not an independent re-grade. Lane FRZ1R built the row blind, before FRZ1's result was opened: FOREST ROW FAILS (8142.08/7685.51), and its comparison reads AGREES WITH DIFFERENCES THAT DO NOT MOVE A LABEL. Both rest on the record's inherited mass-function and concentration ingredients.

Lane FRZ2 let a_0 change with redshift: NO FRAMEWORK $a_0(z)$ PASSES, including every power law. The nearest case, $a_0 \propto (1+z)^{0.5}$, reads 252/167, still 40 to 60 times over the gate; the dark-energy-density cases lower a_0 at $z = 3$ and make the deficit worse. Lane A0Z1's $a_0(\chi)$ also FAILS FRZ2's gate. Lane FZS1 laid the law out

only inside a settled radius, a rule post hoc to FRZ1's fail: FOREST ROW FAILS, FRAGILE, $\Delta\chi^2$ 38.7/36.7, with the half-radius bracket passing.

Lane SRH1 followed one history of the settled radius, f times the fold radius, against the forest, rotation curves and galaxy lensing: WINDOW EMPTY on both footings. The forest needs $f \leq 0.70$ on the grid (edge 0.722/0.725) and KiDS lensing $f \geq 1.25$ (edge 1.004/1.050). Its post hoc amendment, reported and never graded, opens a window with a redshift tilt, the outcome its criteria named in advance as counting against. A re-run from a copy reproduces its label and is not an independent re-grade. The blind re-grade of its lensing gate, SRH1R, reads NO VERDICT on both footings because two of three mutation twins do not bite; its gate passes at every f in the window and also at $f = 0.05$, so lensing cannot tell one f from another. SRH1R does not relabel SRH1. Lane FINAL4 grades the growth history of the formation reading on the same gate: DERIVED HISTORY EXCLUDED on both footings, with FRZ1's $\Delta\chi^2$, and TILT POST HOC, NOT EXCLUDED [PENDING: an uncommitted correction notice in closure_2026_11_11/FINAL4_growth_history/RESULT.md bears on whether this history is derived; the committed wording is kept until it is committed or withdrawn].

8.2 The joint

Rule 6 (Section 7.1) stays underived, and the medium is not identified. Each lane below took one carrier or one route to settling; the table gives the rest.

- **Settling.** SET1, relaxation towards the law's own pressure: STRUCTURE FAILS AT A1 on both footings. For extended baryons the force is off by exactly $a_0\rho_b$, and cored baryons need a negative cold density in the core. AKS1's Part T, itself post hoc, finds that no local dissipation of the clock settles it into the law's halo by today, with one branch closed on an estimate. VSH1 computed that estimate: NO VERDICT as frozen; SETTLES BY SHOCK under a post hoc amendment written after the first run's outputs were read. It was not re-run by anyone else and awaits an independent re-grade; audit A3 grades it SOUND WITH LIMITS and notes that its charge is 1 to 4 % of the law's halo. VSJ1 then asked whether that settled state is the joint: NO VERDICT as frozen; STANDS BUT MISCOUNTED under a post hoc amendment; VSH1's settled state is not the joint.
- **Holding and sourcing the charge.** EDG1: FAILS AT E3 on both footings; a shell of orbits in all orientations can be held, but infall gives it 19 to 38 times too little angular momentum (on a PROVISIONAL spin of accreted matter). VCC1: the charge can be sourced, but every source that can pay for it is homogeneous, so it cannot make the law's halo: EXCLUDED (A3: SOUND). CLOSE2 graded PAR1's two open branches: NO VERDICT as frozen; CANDIDATE STRUCTURE FAILS under post hoc amendments, with the union CLOSED only in the member sense (A2: SOUND WITH LIMITS).
- **The kinetic function of Section 7.2.** CLOSE-1 computed whether a forming halo reaches its static state with that function: FOLDS on both footings (A2: SOUND WITH LIMITS) [PENDING: an uncommitted correction notice in hy4_preview_push/CLOSE-1_DT2/RESULT.md bears on this lane's linear gate; the committed wording is kept until it is committed or withdrawn].
- **Media.** An elastic solid FAILS AT S0 (ELA1; A2: SOUND); the halo's anisotropic stress cannot pay the lensing (STR1); FINAL-7 reads NO MEDIUM, closed at momentum conservation (A2: SOUND WITH LIMITS); a pinned bootstrap carrier is KILLED (OUT5); a contact between cold dark energy and neutral gas is KILLED, one gate PROVISIONAL (NBC1), the class Section 7.3 worked out as arithmetic only. The TOE battery (TOE0) reads NO TOE in either sense: action (A) has no weakly coupled flat vacuum (QG1) and the law's function is no term of a power-counting-renormalizable action (HZ1); a non-perturbative origin gives the shape only (NPJ1), and a Lifshitz fermion superfluid is KILLED AT G0 (LSF1).
- **The formation reading.** FINAL-1 writes the theory out and says it does not derive the joint; A2 grades it OVERSTATED and finds that the formation reading moves the joint into a boundary condition. FRM1 reads HOLDS EXCEPT on both footings and FAILS on its frozen line, a reading that awaits an independent re-grade [PENDING: an uncommitted correction notice in fable52_push/FRM1_formation_theory/RESULT.md bears on this lane's scope; the committed wording is kept until it is committed or withdrawn].

8.3 External field

EFN1 took the nonlinear outcome of SCS3's external-field instability: NO VERDICT on both footings in every reading, frozen and under its four post hoc amendments. Its deciding route stops at a corner of its own boundary, and its second route is not converged (Section 5, item 2).

8.4 Data rows

- **Weak-lensing amplitude.** BX1's second full-size pair finished after version 1.1. As frozen its label is unchanged, NO VERDICT on both footings. Under its post hoc amendment, on the mean of the two pairs, it reads AMPLITUDE CONSISTENT on both footings: KiDS-Legacy +1.46/+1.67 widths and DES Y3 +1.78/+1.76, with DES Y3 alt tagged inside the splice noise and KiDS-1000, reported, at +3.56/+3.69; with the ln-det term DES Y3 stays above 2 on both footings. It awaits an independent re-grade. LWB1, on DESI DR1 lenses, is NO VERDICT as frozen, since a known answer failed in the check, and AMPLITUDE CONSISTENT, FRAGILE under its post hoc amendment: σ_8 $0.786 \pm 0.029/0.787 \pm 0.029$, +0.90/+0.87 widths below form B's neutrino-floor value; it calls itself a consistency check, not independent evidence. Its blind re-grade, LWB1R, passes that check as frozen and so reads AMPLITUDE CONSISTENT, FRAGILE without an amendment (gap +0.91/+0.87); survey by survey KiDS-1000 has NO POWER, DES Y3 is in TENSION (+3.49/+3.47), HSC Y3 is consistent (−1.28/−1.32) and KiDS-1000 with DES Y3 is in TENSION (+3.12/+3.09). Its comparison reads AGREES WITH DIFFERENCES THAT DO NOT MOVE A LABEL. It had seen LWB1's first-run rows before writing code, so it is not a fully blind re-grade, and the two lanes share their data and stored spectra. None of this closes item 12.
- a_0 **at** $z = 2.33$**.** AZ2 is NO VERDICT as frozen and FLAT HOLDS / EXPANSION-MODEL DEPENDENT under its second, post hoc amendment, conditional on Ω_m near 0.30; the lane says FLAT HOLDS is not a pass, since ΛCDM's constant dark energy gives the same, and its lead's re-run does not reproduce byte for byte. Its blind re-grade, AZ2R, reads FLAT HOLDS / EXPANSION-MODEL DEPENDENT on both footings: the headline cell holds (alt $\sqrt{X}$ 1.322), and the flat reading FAILS with BAO and supernovae at Ω_m 0.25 and 0.35, which is why the label is expansion-model dependent. Its comparison with AZ2's amended reading reads AGREES WITH DIFFERENCES THAT DO NOT MOVE A LABEL; its continuous profile passes AZ2's frozen known answer as written, and it does not relabel AZ2's frozen NO VERDICT. It also says that a FLAT HOLDS is not a pass, since constant dark energy gives the same; that its new DESI pair is a PROVISIONAL reading of the source; that it shares AZ2's data, model and criteria; and that AZ2's amendment files, readable before its blind commit, already quoted the headline numbers. It closes no open item.
- **The Milky Way.** DW1's frozen criteria award 9 of 12 checks; the extrapolated M_{200c} fails. NMB1: the line PASSES the shape-measuring DESI curves in all four cases, FRAGILE in one. PLT1: PASSES, FRAGILE with the Local Group field and PASSES without one; on a pre-named variant beyond 150 kpc the satellites FAIL LOW (median −2.72). MWG1: hot gas does not supply the rotation curve (ROTATION CURVE NOT SUPPLIED), and with hot gas counted the outer rows pass, FRAGILE, on PROVISIONAL gas masses. MWJ1 is a post hoc consolidation with every input seen and reads CLOSES WITH CENSUS BARYONS, WEAKER FIELD, ROTATION CURVE INSIDE. Each pass awaits an independent re-grade.
- **Euclid strong lenses.** These grade the law acting on the lens galaxy's stars (law S). EUC1, on 3,192 lenses with every source at infinity: NO VERDICT on both footings, one-sided; the median lens sits at the ceiling. EUC4, on 124 wide lenses: WIDE LENSES BREAK THE CEILING on both footings, 75.8 % and 70.2 % above 1.78 times Salpeter; its blind re-grade EUC4R gives the same word and says its numbers are blind and its word is not. EUC2 (source redshifts from DESI fibres) and EUC5 (dynamical boost) are NOT EVALUABLE as frozen and under their amendments.
- **The amount.** GAS1: MISSING BARYONS EXCLUDED [AMOUNT AND AMPLITUDE] on both footings (A3: SOUND WITH LIMITS). CLS2 and GO2 are in Section 5. U4L1, a secondary reading of an ALMA limit with no verdict weight, has NO LABEL as frozen and NO TENSION under its amendment; one disc excludes no model.

8.5 κ and the share

No lane derives κ or 5.41. FINAL16, FINAL-17, FINAL18, FINAL-19, FINAL6, OUT4 and OUT3's scale-free boundary each read KAPPA NOT DERIVED (FITTED) in their own words, and OUT3's cosmic-share fold reads AMOUNT NOT DERIVED (UNAVAILABLE), its construction missing by 18.5 band-widths. OUT-2's headline is that a clock-rate lock makes κ an output, $\sqrt{2/(3\pi)} = 0.460659$; audit A1 grades it WRONG: postulating the lock is algebraically the same as postulating that value of κ, so nothing is derived. A1 grades FINAL16 and FINAL-17 OVERSTATED (their verdict stands; their claims of criteria committed alone are false, and FINAL-17's capacity is one line of the law's algebra), FINAL18, FINAL-19 and OUT3's fold SOUND WITH LIMITS, and OUT4 SOUND. The Lean lanes certify arithmetic in which 5.41 is an input (FINAL-11, FINAL-14) or do not compile (FINAL-10); FINAL-12's committed result reports a compile at exit 0, and A1 records that the copy it built is

untracked, differs in its bytes and fails. Of the formation lanes, FINAL2 reads NO POWER, FINAL15 FAILS as frozen and holds only after a correction of its own specification, FINAL7F reads NO STRUCTURE as frozen, and FINAL6 reads KAPPA NOT DERIVED (FITTED) + FORMATION REFUTED (per-galaxy) [PENDING: an uncommitted correction notice in closure_2026_11_11/FINAL6_kappa_formation/RESULT.md bears on this label's scope; the committed wording is kept until it is committed or withdrawn]; A2 grades FINAL15 and FINAL7F OVERSTATED, and FINAL15's holding reading awaits an independent re-grade. FINAL5 proposes a stacked thermal SZ measurement and grades nothing; A2 finds it not decisive for the joint.

8.6 Every lane folded into version 1.2

Labels canonical / alt as each file states them; "both" means the same label on both footings. "awaits" means awaits an independent re-grade.

Lane	Question	Label as frozen	Amendments (post hoc)	Re-run, blind re-grade, audit
The forest				
FRZ1	forest row with the redistribution at z 2.2–4.4	FOREST ROW FAILS / FOREST ROW FAILS	none	lead's re-run reproduces; blind build FRZ1R
FRZ1R	blind build of FRZ1's row	FOREST ROW FAILS (8142.08) / FAILS (7685.51)	none	comparison: agrees
FRZ2	forest with $a_0(z)$	NO FRAMEWORK $a_0(z)$ PASSES	none	
FZS1	law's layout only inside the settled radius	FOREST ROW FAILS, FRAGILE (both)	the rule is post hoc to FRZ1	
SRH1	one settled-radius history, forest against lensing	WINDOW EMPTY / WINDOW EMPTY	A1: tilt cases, reported, never graded	re-run reproduces; lensing gate re-graded blind (SRH1R)
SRH1R	blind re-grade of SRH1's lensing gate	NO VERDICT / NO VERDICT	one, before data	does not relabel SRH1
HLF1	CO1's rows on the own halo list; forest row	NO VERDICT / NO VERDICT	A1: Part A ROWS HOLD WITH THE OWN LIST	awaits
FINAL4	growth history of the formation reading	DERIVED HISTORY EXCLUDED (both)	tilt reading post hoc	
The joint and the medium				
SET1	irreversible settling to the law's pressure	STRUCTURE FAILS AT A1 (both)	none	
EDG1	charge held in an edge layer	FAILS AT E3 / FAILS AT E3	none	
AKS1	acceleration-keyed sink; local dissipation	control as recorded; Part T CLOSED, ONE BRANCH ON AN ESTIMATE (both)	Part T is post hoc; A1	
VSH1	AKS1's estimate T7, computed	NO VERDICT / NO VERDICT	A1: SETTLES BY SHOCK (both)	not re-run; awaits; A3 SOUND WITH LIMITS
VSJ1	is VSH1's settled state the joint	NO VERDICT / NO VERDICT	A1: STANDS BUT MISCOUNTED (both)	not re-run
VCC1	can the clock's charge be sourced	EXCLUDED / EXCLUDED	none	A3 SOUND
CLOSE2	PAR1's two open branches	NO VERDICT / NO VERDICT	CANDIDATE STRUCTURE FAILS (both)	A2 SOUND WITH LIMITS
CLOSE-1	engineered kinetic function	FOLDS / FOLDS	none	A2 SOUND WITH LIMITS
PIK1	dust term gated on the halo density	NO VALUE / NO VALUE	none	a re-run is owed (its file)
A0Z1	the χ sector and $a_0(\chi)$	EOM CERTIFIED; $a_0(\chi)$ FAILS (both)	none	
FRM1	the formation reading as a theory	frozen line FAILS / FAILS; reported HOLDS EXCEPT (both)	one, before numbers	awaits
FINAL-1	the theory assembled	THEORY WRITTEN (both)	none	A2 OVERSTATED
FINAL-7	what the medium is	NO MEDIUM, CLOSED AT (ii) (both)	none	A2 SOUND WITH LIMITS
ELA1	an elastic solid as the medium	ELASTIC SOLID FAILS AT S0 (+S2, B1) (both)	A1	A2 SOUND
STR1	can the anisotropic stress pay the lensing	STRESS CANNOT PAY (both)	none	A3 SOUND WITH LIMITS

Lane	Question	Label as frozen	Amendments (post hoc)	Re-run, blind re-grade, audit
OUT5	a pinned bootstrap carrier	BOOTSTRAP KILLED (FRW/HALO + PAYMENT)	none	A2 SOUND WITH LIMITS
NBC1	contact with neutral gas	KILLED (K2 PROVISIONAL) (both)	none	
TOE0	gate battery for a theory of everything	NO TOE, in either sense	A1, before any number	
QG1	quantum health of action (A)	T2 FAIL: Minkowski FAILS (both)	A1, before the graded run	
HZ1	a UV completion of action (A)	T3 FAIL	A1 (one MUTATE twin)	
NPJ1	a non-perturbative origin of J	SHAPE ONLY	none	
LSF1	a Lifshitz fermion superfluid	KILLED AT G0 (both)	none	
External field				
EFN1	nonlinear outcome of SCS3's instability	NO VERDICT / NO VERDICT	A1–A4: NO VERDICT	
Data rows				
BX1	pair B finished (addendum)	NO VERDICT / NO VERDICT (unchanged)	A1: AMPLITUDE CONSISTENT (both)	awaits
LWB1	amplitude on DESI DR1 lenses	NO VERDICT / NO VERDICT	A1: AMPLITUDE CONSISTENT, FRAGILE	blind re-grade LWB1R
LWB1R	blind re-grade of LWB1	AMPLITUDE CONSISTENT, FRAGILE (both)	none	not fully blind (its file)
AZ2	flat a_0 at z 2.33, DESI DR2 Lyα	NO VERDICT / NO VERDICT	A2: FLAT HOLDS / EXPANSION-MODEL DEPENDENT	re-run not byte-identical; blind re-grade AZ2R
AZ2R	blind re-grade of AZ2	FLAT HOLDS / EXPANSION-MODEL DEPENDENT (both)	none	blindness caveat (its file)
GO2	group census on DESI groups	NOT EVALUABLE [r_ta NOT REACHED] (both)	none	lead's re-run reproduces
CLS2	what the cluster mass is	NO CANDIDATE / NO CANDIDATE	none	
CRACK-1	three readings of the ultra-faint and X-COP pair	FAILS THE PAIR / FAILS THE PAIR	A1; same labels	A3 SOUND WITH LIMITS
GAS1	is the amount missing baryons	MISSING BARYONS EXCLUDED (both)	none	A3 SOUND WITH LIMITS
DW1	DESI DR2 Milky Way mass profile	9 of 12 checks; M_{200c} fails	none	
NMB1	line against DESI's NIMBLE curves	PASSES (E1 canonical FRAGILE)	none	awaits
PLT1	outer masses at 100 kpc and beyond	PASSES, FRAGILE (field); PASSES (no field)	a post hoc reading	awaits
MWG1	hot halo gas	ROTATION CURVE NOT SUPPLIED; outer rows PASS, FRAGILE	none	awaits
MWJ1	one Milky Way window	CLOSES WITH CENSUS BARYONS, ...	post hoc consolidation	awaits
EUC1	Euclid lenses at the ceiling	NO VERDICT / NO VERDICT	none	
EUC2	source redshifts in DESI fibres	NOT EVALUABLE	A1: NOT EVALUABLE	
EUC4	wide Euclid lenses	WIDE LENSES BREAK THE CEILING (both)	none	blind re-grade EUC4R: same
EUC4R	blind re-grade of EUC4	WIDE LENSES BREAK THE CEILING (both)	none	word not blind (its file)
EUC5	dynamical boost of lens galaxies	NOT EVALUABLE / NOT EVALUABLE	A1: NOT EVALUABLE	
U4L1	ALMA limit on one disc	NO LABEL	A1: NO TENSION	no verdict weight
κ and the share				
FINAL16	is κ an output	KAPPA NOT DERIVED (FITTED), FORM OUTPUT	A1 (J7), before the run	A1 OVERSTATED
FINAL-17	κ from the capacity	KAPPA NOT DERIVED (FITTED), CAPACITY DERIVED	A1, after the outputs (audit A1)	A1 OVERSTATED
FINAL18	κ from matching at the fold	KAPPA NOT DERIVED (FITTED), FOLD MATCHING FALSIFIED	none	A1 SOUND WITH LIMITS

Lane	Question	Label as frozen	Amendments (post hoc)	Re-run, blind re-grade, audit
FINAL-19	κ from the deep-field column	not derived; the column is 6.0/7.3 times too large	none	A1 SOUND WITH LIMITS
FINAL6	κ from the formation reading	KAPPA NOT DERIVED (FITTED) + FORMATION REFUTED	none	
OUT-2	a clock-rate lock	$\kappa = 0.460659$ claimed as an output	none	A1 WRONG
OUT3 (fold)	the amount from the fold	AMOUNT NOT DERIVED (UNAVAILABLE)	none	A1 SOUND WITH LIMITS
OUT3 (boundary)	$\kappa = \frac{1}{2}$ and the vacuum value	NOT DERIVED (FITTED)	A1, before the Lean run	A2 SOUND WITH LIMITS
OUT4	the κ window	KAPPA NOT DERIVED (FITTED); WINDOW 9.2x TOO WIDE	none	A1 SOUND
FINAL2	which epoch fixes the amount	(e) BEST-RANKED / NO POWER (both)	none	
FINAL15	the formation ledger	FAILS / FAILS	specification defect D1 corrected: HOLDS / HOLDS IN PATTERN, NOT IN SIZE	A2 OVERSTATED; awaits
FINAL7F	residual as history or present state	NO STRUCTURE / NO STRUCTURE	A1: NOT A FUNCTION OF ANY TESTED VARIABLE	A2 OVERSTATED
FINAL5	the deciding observation	a proposal, no label	none	A2 OVERSTATED
FINAL-10/11/12/14	Lean certificates	FAIL; CERTIFIED; exit 0 (30 theorems); six items compile	none	A1: see text

9 Our own estimates

Recorded before the numbers that tested them. (i) The KO2 brief put the error on the disc ratio near 0.2 and expected discs low and early types at 0.7 to 1.0. A model-only toy at the freeze gave the brief's ratio about twice the error of its replacement (0.37 against 0.19), so the lane replaced the primary statistic before computing; the brief's ratio then measured an error of 0.278 on discs, and discs came out undecided. (ii) The expectation that discs carry no outer field rested on the host-free lead, which does not generalise to the full sample. (iii) The RB1 brief expected X-COP near 0.96; it reads 1.095 on the forward-fitted mass. (iv) RB2 expected M31's bright dwarfs to be borderline; they fail. (v) The Bullet estimate, closer than 2.6σ once the stripped gas is counted, held; so did the group range and the Coma shift. (vi) CO2's hand estimate, written into its criteria before any code, put the optical depth at 0.055 to 0.060, σ_8 near 0.828, PR4 lensing within 1σ and raw KiDS near the line between tension and exclusion. The optical depth came out just below that range (0.0532); the rest held. (vii) The lead session and TPA1 expected the instability to survive the coupling to gravity, and the draft of this note carried it as the static state's own. It did not survive: SCS1's expectation before its freeze, stable with the earlier result planar, is the outcome. Version 1.1: (viii) The closure session expected the survey's newer release to cut the weak-lensing gap from about five widths to one or two, and the neutrino floor, the covariance and the gas together to be worth about one more. The release cut the σ_8 gap by about 45 % and left 4.4/4.6 of Legacy's narrower widths; the three inputs were worth 2.3 more widths on Legacy and 1.7 on KiDS-1000. (ix) It expected gas from the stellar-mass-selected stack to be worth 0.5 to 1 widths; it was worth 0.05 to 0.09 and only a bracket. (x) It proposed that the dwarfs that sit high are the ones that lost their gas. DG1 found the class medians matched and the per-galaxy gates failed. (xi) It proposed the acceleration key and wrote that its gradient term had a healthy sign. AKD1 found the sign the other way: the term is kinetic, with a negative coefficient. Version 1.2: (xii) FRZ1's hand estimate expected the forest row to fail, possibly FRAGILE; the redistribution came out with the opposite sign and ten times the estimated size at k 3 to $5\,h\,\mathrm{Mpc}^{-1}$, and the fail is not FRAGILE. (xiii) FRZ1R's hand estimate, written blind, also had the sign wrong (+0.02 to +0.10 expected). (xiv) BX1's estimate before its second pair called GAP STAYS on DES Y3 most likely; it was not reached, and the measured noise came out below the estimated range. (xv) ELA1's hand estimate had a sign wrong that changes the outcome of the gate expected most likely to kill; its amendment, post hoc, corrects it. (xvi) PLT1 expected the halo stars and the satellites to split; they did not.

Method

Lanes KO2, RB1, RB2, CO1 and CO2 are under `closure_2026_11_09/`; HE1, ONE1, TPA1, SCS1 and the notes FB1 and KO3F are under `fable52_push/`. None is registered in the closure record and none relabels an earlier lane. Post hoc items are marked in their lanes: RB1's amendment on the cluster shape gate, KO2's four amendments (the third, form B's prediction, was committed before any fitted amplitude was opened), KO2's mutation twin that did not bite, CO1's restated mutation arm, CO2's tolerance amendment, CO2's realisation check, which was made after grading, and SCS1's three amendments. RB2 has no amendment. This note claims no Lean theorem of its own; the Lean build that earlier lanes compiled with was removed from the machine on the evening of 2026-11-09 and a build inside the repository has been used since 22:17 that evening, so a Lean file's status is what its lane's compile output says. Distances for the KiDS lenses use the record's fiducial flat background (Ω_m 0.3, H_0 70), as in the staged sums. No halo model, no concentration relation and no simulation fit enters a graded row of KO2, RB1 or RB2; one externally calibrated bracket in RB1 is labelled as such. CO1's redistribution rows, which CO2's weak-lensing row reuses, inherit a mass function, a bias and a concentration relation calibrated on simulations, and say so. CO2's low-multipole table is Planck 2018 and its high-multipole spectra are PR4. The owner's decision is recorded in `closure_2026_11_09/OWNER_DECISION_2026-11-09_FINAL_FORM_B.md`. The supplement holds every lane file.

Version 1.1's lanes: SCS2, SCS3, DCN1, MK1, MK2, COL1, FLD1, REL1 and SHP1 are under `fable52_push/`; TK1, AKD1, LFS1, CLS1, KL1, KL1R, DG1, NH2, XCV1, GO1, BX1, DSB1, PAR1, WV1, SMF1, the note R541 and its lanes R541A to R541E are under `closure_2026_11_10/`, beside the notes `THEORY_ASSEMBLED_2026-11-10.md` and `MEDIUM_DESIGN_2026-11-10.md`, which are assemblies and grade nothing. None is registered and none relabels an earlier lane. Post hoc items are marked in their lanes: KL1's amendments A1, C1, C2 and C3, on which its readings other than NO VERDICT rest; DG1's amendment; TK1's, FLD1's and REL1's amendments; COL1's four amendments; MK2's second, third and fourth amendments; SCS2's second to fourth, SCS3's four and DCN1's six; GO1's amendment; PAR1's two amendments, WV1's two and SMF1's one; DSB1's amendment, made after its freeze and before any graded number; BX1's amendment; SHP1's check made after grading; and the readings of NH2 and XCV1 other than NO VERDICT. WM1's seeding gate was added after its first result. AKD1 has no amendment; its extra readings were added after the freeze and set no label. The earlier lanes named in Section 7's survey of media (M5, PH1, IJ1, HF1 and TM1 under `rebuild_2026_10_22/`, QX28, FA3 and WM1 under the closure folders of 2026-11-06, 11-07 and 11-09) are quoted by the first lines of their result files only; those files are added to the supplement. The KiDS-Legacy release was fetched on the owner's go and sits beside the repository, not in it.

Version 1.2's lanes: CLS2, EDG1, EFN1, FRZ1, FRZ1R, FRZ2, FZS1, HLF1, SET1, SRH1, SRH1R and VSH1 are under `closure_2026_11_10/`; DW1, FINAL-12, FINAL15, FINAL16, FINAL18, FINAL2, FINAL4, FINAL6 and VCC1 under `closure_2026_11_11/`; AZ2, AZ2R, CLOSE2, EUC1, EUC2, EUC4, EUC4R, EUC5, FINAL7F, GO2, LWB1, LWB1R, MWG1, MWJ1, NMB1, OUT4, PLT1 and U4L1 under `closure_2026_11_12/`; A0Z1, FRM1, NBC1 and PIK1 under `fable52_push/`; CLOSE-1, CRACK-1, ELA1, FINAL-1, FINAL-7, FINAL-10, FINAL-11, FINAL-14, FINAL-17, FINAL-19, FINAL5, GAS1, OUT-2, both OUT3 lanes and OUT5 under `hy4_preview_push/`; STR1 under `hy4_push/`; AKS1, VSJ1, TOE0, QG1, HZ1, NPJ1 and LSF1 under `toe_2026_11_10/`. The audit quoted beside them is `reviews/hy4_codex_audit_2026_11_12/` (A1 to A4). None is registered and none relabels an earlier lane. Post hoc items are marked in their lanes: HLF1's A1; FZS1's rule and SRH1's family, both chosen after FRZ1 was seen, and SRH1's A1; SRH1R's A1; VSH1's and VSJ1's A1; AKS1's Part T and its A1; EFN1's A1 to A4; CLOSE2's amendments; FINAL7F's A1; FINAL15's correction of defect D1; FINAL-17's A1; OUT3's A1; ELA1's A1; CRACK-1's A1; BX1's A1 and its addendum; LWB1's A1; AZ2's A1 and A2; EUC2's and EUC5's A1 and EUC5's calibration; U4L1's A1; PLT1's post hoc reading; MWJ1, a consolidation made with every input seen; TOE0's and QG1's amendments; and HZ1's added twin. The forest lanes rest on the record's inherited mass-function and concentration ingredients, and FRZ1's lead did not check their scalings against their sources. The running lane FRZ3 is marked pending where its sentence will go.